

Chris Frith ~ Active Inference Insights 009 ~

Mentalizing, Schizophrenia, Consciousness

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hello everyone and welcome back to active inference insights I your host darus parvey Wayne and today I absolutely thrilled to be speaking with Professor Chris FFF Chris is a neuropsychologist and a Pioneer in the application of brain Imaging to the study of mental processes he has contributed more than 500 papers to peer-reviewed Scientific journals and is known especially for his work on agency schizophrenia and social cognition within this latter he has worked extens extensively with his wife Professor utaff together they have published seminal papers on theory of mind and social signaling and last year they published the book what makes us social again it is no overstatement to say that Chris possesses a legendary status in the Spheres of Psychology and Neuroscience and so it is a real Honor on my part to be chatting to him today Chris thank you so much for joining me and welcome to the show oh thanks for those nice words you're welcome uh it's are you are you in London currently it's a beautiful day yes all be a bit Frosty yeah uh excellent so as I mentioned in my introduction uh you root many or you ground many of our social capacities in mentalizing which requires theory of mind now some of our audience might not we we our audience is diverse you know act brings in people from maths physics all sorts they might not have a psychology background so they might not be aware of what mentalizing or theory of Mind actually is so it might be worth just setting the stage and giving a brief exposition of what those terms mean if possible right oh yes theory of mind is a rather misleading phrase because it's not a theory of how the mind works and it's probably not even a theory so mentalizing is a slightly better term I think but the idea is that a one way into this is to say we're trying to predict prct is useful to be able to predict what people are going to do and to understand why they're doing it and theory of Mind originally said people do things because of their hidden mental state such as beliefs and intentions so if somebody takes an umbrella with them this doesn't mean that it's raining it means that they believe it is reigning or is going to rain and the key um to the experiments on people's ability to understand these hidden mental States is the impact of false

beliefs because if somebody if you know it's not raining and this person goes out with an umbrella then you can say he has a false belief that it's going to rain and his belief is different from mine and I have to take this into account when I predict what he is going to do so that's basically what mentalizing is about that's the process of predicting what other people's mental states are and mental States meaning things like intentions beliefs and desires and we think that this is probably different from understanding people's emotions yes I can imagine a lot of people make that conflation um I guess and a further thing to unpack there is when people hear about sort of mentalizing they might think of autism and the associated difficulties in mentalizing that seem to be associated with autism but over time uh many autistic people use heuristics or tricks or social cues to mentalize yes is there a difference because behaviorally they might manifest similarly and permit kind of harmonious social interaction or is theory of mind or mentalizing committed to saying that uh those who are mentalizing actually have a mental representation a kind of recursive mental representation of what the other part in this thinking um well that's what I think I don't think everybody agrees with that there is I mean more recently there is evidence that there are two I mean depends on how you decide to talk about it two levels of mentalizing which some people say the low one is not really mentalizing but that is to say some sometimes called Behavior reading um that you can solve many you can make many of these predictions without needing to know about other people's mental States and um this is and it's so all also in some sense automatic so it happens you don't have to think about it you just um know this is the right way to behave next in in this interaction and in relation to autism there's some suggestion that it's this lower level part that they find difficult and that the heuristics as you call them are in fact something that we all use so there's this there's this thing called folk psychology which is supposed to be inferior but I think is actually very important which is that we know this is the idea that people do indeed behave because of their beliefs and desires and that's how we can predict and their behavior and this is a cultural phenomena so in different cultures you may have different ideas about how folk psychology works and in particular in um there's some anthropological work on the C in people who come from the Pacific Rim which includes some Meers in Mexico and Fiji and these various other places where they there's a cultural attitude that it's impossible to know what other people's mental states are and that you shouldn't bother to try and in fact children are brought up not you know you shouldn't be doing this sort of thing you just look at their behavior and it even extends to sort of interesting semi-legal situations where in the west we're very concerned that you know it's the intention that counts not it's more important the intention

is more important than the outcome whereas in these other cultures the outcome is all that matters and even if it's an accident you'll be punished interesting there's an interesting level of I think cultural effects on some of these mentalizing stories yes philosophically I certainly have issues with that but we won't we won't go into that too much um yes that's interesting I guess because it also maps onto our scientific institutions and our way of actually viewing what's going on mentally um EXA I can't help but notice that we are presupposing mental representations even when I asked that question and I've been sort of delving into the world of radical inactivist cognition and they would say well there are no representations there are there is no content all we have are these Dynamic relationships between or these Dynamic couplings uh between agent and Arena right does that is that does theory of Mind collapse if we just permit that these are um I guess they talk about sort of perceiving correlations in the environment that's something that they're very excited about but actually this isn't a in you know this isn't necessarily a representation or intentional stance um does it does it is it compatible with these sort of ramped up versions of an activist cognition hello everyone and I do apologize for interrupting the flow of the conversation with Chris I wanted to add this little bit because I realized uh upon watching the conversation that I had with Chris that this juncture I did misspeak and misconstrue and overgeneralize the claims of wreck at least explicated by hat and Mayan so I made it out as if uh they say that all cognition invol uh is contentless this is not actually what they say they say that basic mind or basic cognition is contentless whereby basic cognition means all cognitive activities except those involving public Language and Cultural symbol systems so again they do not say that all cognition is contentless because they because there are also these higher order forms of cognition which might be content involving and again those require public symbols social practices of symbolic uh communication so they're therefore governed by semantic norms and these correctness conditions so where the theory of mind is that uh that higher order form or lower order form I guess is for you to decide but you'll see later on I speak about these kind of scaffolded uh developmental capacities which might involve content as if it wasn't part of the theory of heart to mind but it is so I really wanted to clarify that before this goes uh before this podcast goes live again uh just more broadly I do want to take responsibility as your podcast host for saying things are accurate and truthful and owning up I make mistakes this was one of those uh places I would have edited out but it really would have messed up the flow of the conversation so I thought it was easier for me to just come on here and clarify that I made a mistake and hopefully uh yeah clear up any confusion again um I don't mind too much if I

mess up describing what I think because they're my thoughts but I know that authors far more senior than me have put a lot of time and a lot of effort into their accounts and so it'll be doing a disservice to them disservice to you if I misconstrue or misspeak when describing uh their for and their writings so when I do this again because you know In the Heat of the Moment In the thir of the moment I will make mistakes as I did here and again it's just because these are live conversation so I'm having to think on my feet uh but when I do it again because unfortunately I am fallible please do tell me uh I will always look out for it I'm happy that I spotted this one but again I feel a very strong responsibility to be as accurate as possible so that relies on you as well so please don't inform me for when I make a mistake and I will not keep you any longer because this is a cracking conversation with Chris and I hope you enjoy the rest of it yeah I'm not very keen on these ramped up versions um I would it's conceivable that this might be useful way of looking at the sort of automatic mentalizing but I think the sort of cultural um folk psychology story very much depends on representations I mean we almost we're talking about them as it I mean yeah um I can't there's a philosopher called Davidson I can't remember precisely what he said but it's all about applying you know rational arguments to what we believe are these mental States so you're saying if he believes that the chocolate is in the cupboard he will go and look in the cupboard is the sort of um philosophical approach to this yeah I mean and that I think yeah no as you say I mean these are uh sort of an activist cognition on steroids um there of course people and I don't think you would disagree with them that recognize that if there are representational schema those are built off scaffolds that occur from social cognition developmental psychology being with others and I don't think you would disagree with that and that is something I want to touch upon um in terms of self-modeling and self-perception but yeah um actually I'll backtrack I don't think they would say that uh cognition is non-intentional just that it's non contentful or non represent ation um but there are obviously quite severe critiques of that and there probably are representation as well as they make clear in their works so now expanding that out to away from dactic or not dactic diadic interactions yeah into broader social contexts why going Beyond just the coupled Duo why is theory of mind or mentalizing so important just for social institutions more broadly well I I I I would claim that we don't think that necessarily okay so we think that that some people seem to equate social cognition with theory of mind right and we do not sure okay let's clarify that yeah so we think there are all sorts of social cognitive processes that are not to do with representing mental States or whatever and many of those might be more relevant to larger group interactions and we talk about I mean what's and the original paper that we wrote

in 1999 I think which is sort of starting of social cognition we had mentalizing on one side and we had mirror neurons on the other side because they had only recently been confirmed and um mirans are all about how we can imitate others and how we get um what's the word contagion from others so for example emotions are contagious we tend to feel the emotions that other people are expressing we tend to do what other people do um we imitate their actions and one very important aspect of human sociality is overimitation do you know about yes I I I've seen some of it in sort of uh linguistic uh Innovation where a language learner learning another's language will overimitate and so they'll be sort of uh uh they'll all speak in RP for example even though no one's spoken in RP since 1910 yes but I'm as I've not thought about it so much in that respect but it's more to do with simple actions um like shaking hands is and you have to do it in the right way and the early experiments where they had these um puzzle boxes which you have to solve by going through series of Maneuvers and the trainer puts in a couple of Maneuvers are not actually necessary to open the box but the children all imitate these as part of the that's the overimitation bit so that they do them even though they're not I see I see and the interesting thing is that chimpanzees don't very interesting the idea is this is we're imitating not to get to the reward we're imitating to show that we are doing it properly we are part of the group we are the people who behave like this and they're yes yes so it's kind of in a sense directed The what in active inference terms kind of a higher a higher order prior yes no that's that's very interesting yeah I mean um I guess to return to my initial question yes I I recognize it might be an overextension of theory of mind I didn't mean it in terms of everything is Fe of mind that's clearly not the case because then it would be sort of serious cognitive overload if I was just mentalizing everyone and everything and it sounds borderline schizophrenic to be thinking that you know my chair has intentions but it seems to me that even in making or or or acting to confirm that higher order prior that I am like you and so you're going to like me and and and we're going to be able to establish this recursive relationship I need to have at the very fundament of that a notion that you are an intentional conscious agent which which kind of seems to me to be a very lowlevel form of theory of mind it might be implicit in our interactions but that still seems to be present yes that's a very interesting idea um one of the things that we've written about is sort of addressing this question how do we know when we're wanting to in competitive situations and which usually means you're playing an economic game like hide and seek yeah um you need to know what sort of agent you're competing against sure and um if it's a very you know it could be a completely random agent that just is biased so it's very easy to beat one of those or it could be a simple learning agent

that depends on what's you know the pattern reward in the past and it's fairly easy to beat one of those and the most difficult ones to beat are the ones that are intentional and are trying to predict what you're going to do as well as at the same time as you're trying to predict what they're going to do like rock paper scissors or or something like that yes and then that's where you get into this in infinite recursion where you say does he have a representation of what I think he's going to do next yeah yeah yeah yeah yeah and um what is interesting at least the little bits of information I have it's very difficult to work out that you're competing against an intentional agent from their behavior and therefore we tend to depend on our prior if we think it's intentional agent will behave in such and such a way if we think it's not we won't and there's this nice experience John donzo's group in Paris where he has people he has artificial AG that can do recursive Tom yeah and he has people playing against them you know at various levels you can have zero Tom one Tom two Tom and so on in terms of recursion and if you tell them they're playing against a one armed Bandit they assume that it's not intentional and they lose when the agent is functioning at two two levels of recursion even though they have several hundred they think it's yes if they think it's a person they will adopt the high level strategy and of course this is reasonable because this high level of mentalizing is extremely cognitively um demanding yes so you're going to avoid going to that level if can yes and there's this nice and do you know about the beauty contest game which is a nice example of this I don't ah this is invented by Jean mayard KES anyway the task is you're part of a group everybody has to produce a number between one and 100 and the person who produces half the mean production will win okay and it ke Kane's call it a beauty contest game because when you have to say who is the most beautiful person it's not who is the most beautiful person but who people think is the most beautiful person anyway the argument is you know if if you think about what are the other you have to think about what are the other people going to do and you think if they're completely stupid then this will be a 50 will be the average number and therefore you'll win with 25 if they're not completely stupid they think it's going to be 25 therefore you'll win with 12 and a half if you've read about um Nash equilibrium you think everybody is super rational and you're say one so you always lose um and the average I mean there have been huge experiments on this in newspapers where they get thousands of people to take part and the average number level of recursion is two and a half interesting yes it has to be a cut off there has to be a cut off this is what they call I think car would call it bounded rationality or something yes yeah yeah well yes I guess it's also I guess in some sense to do with the depth of your temporal hierarchy yeah um because each recursive step seems to

me at least to be kind of an N plus one in your kind of information space and so you're just going to the next one and it has to be an end at some point we can't plan forever that's fascinating okay that's fascinating yeah no that's that's the other game I'm interested in is the um stank Hunt game you know that yeah I know the fact yeah yeah Y and that's again seems to be recursive because if I go to hunt STS I have to believe that you're going to hunt STS and I have to believe that you think that I'm going to hunt STS and it seems to me theistic there that soles all this to say we are Cooperative people and I don't have to worry about this yeah yeah a couple things come to mind there I mean dennet as I'm sure you know has this notion of an intentional stance um you think that maps ontologically onto our sort of priors in some sense that we we start off with the notion that things are intentional animate conscious however you want to term it and then we start pruning or is it the other way I think I think it's the other way around we want to start with the idea that it's all very easy yes yeah yes but but we do have I I think we have a prayer that people are intentional and and would that be and would we get that evidence from the phenotype of people so to speak so so if you have the sim a similar body to me you move in a similar way to me you speak in a similar way to me that is evidence to confirm that prior but I guess it has to be an evolutionary prior quite a non conceptual level because Evolution doesn't work like that it's not going to say if you've got if you're six foot you're an agent right it's it needs to be a little bit more subtle than that yeah but certainly there's this um somewhat discouraging work on ingroups and out groups groups are considered to be less intentional gosh yes that's quite terrifying yeah yeah yeah but that said I mean we do have it makes me think of the Fus of form facial area and the fact that that certainly constitutes some kind of perceptual prior at least yeah excellent that's very good um I mean I was listening to an interview with that you and UA did with uh I think it was the denter it would have been about a month ago and I think I I just was listening to the beginning and UTA I think it was UT that we could consider cognition all like entirely social cognition so all cognition is social cognition it makes me think my Master's feis was uh exploring what's called social Baseline Theory and I wonder what you thought about it I'm not sure you're aware of social Baseline Theory I'm not so this is um a kind of the experimental work for it started in of 2006 2007 and then the theory was fleshed out sort of early last decades this people like James Cohen I think it's Cohen it might be Cohen um Elizabeth Gross these are people that the audience can search up but the notion is is that the primary ecological niche we find ourselves in is social yes and so what that means is that the human brain for adaptive reasons expects social resources because it and in doing so it that acts to mitigate risk and also

preserve energy expenditure yes so what they then say in white it's called social Baseline theory is that being with others constitutes a cognitive Baseline so that when we actually put people in an fmri scanner we get them to do a task alone which is the standard in cognitive psychology we're actually viewing them as a deficit because they have to engage in all of this self-regulation self-regulatory processes that they don't they would be doing to a lesser degree if they were have trusted others how does that sound to you as a theory oh I I I I love that I'm very sad that I didn't know about it it's really it's there's this really the really beautiful experimental studies I think the most famous one is that people will perceive a hill to be less steep if they're with someone else oh yeah because they I think the perception in some ways again this feeds into a broader question about to what degree is perception influenced by top-down factors is it modular I mean I come from active influence I'm saying no like it's definitely influenced but I think notion is a perceptual uh yeah a percept is influenced by your notion of how much effort it is going to be in some sense the affordance how much effort is going to be to exploit that affordance so being with others reduces that effort no I think that I would entirely agree with that and there's other experiments that we did talk about in the book which I didn't know how they relate to this and I'm not sure whether the Baseline people know about this about affordance this is this business if you if you have a table with mugs on it you will you can show that you will get when the mug appears as it were if it's within your reach you will get um motor activity in motor cortex and there's a social version of this if there's a second person sitting at the table even a a mug that's out of your reach but in his reach yeah Will activate the motor system so that you're treating this is sort of group affordance it's I call it the wi mode it's within our reach yeah and that seems to happen automatically so that's entirely consistent with this that is that is wonderful I guess in some sense from an active inference perspective it's worth predicting others actions as if they were our own because their actions are so influential on on personal outcomes but I was thinking more it's like the monkey and the rake we can use this person as a tool to get yes yeah ah yes ah okay so mine was a bit more generous I guess it was a little bit more altruistic ah okay okay cool yeah that's that that's that's an interesting um that's an interesting stance on it I guess I mean the answer yeah a test of that would be if the P other person sitting at the table is your enemy would would you then not get the affordance from the things yeah has someone done that experiment I don't think so no worthwhile um an interesting experiment thought experiment that I think you've used and Carl has used is let's say we're on some exoplanet and it's just me yeah yeah um I don't I I want to leave the questions about agency and sense of self for a bit because I do want to get

there that's kind of more my speciality the work that I focus on is more to do of self modeling but we say okay all cognition is social cognition it's built into us it's it's our Baseline we have social priors how what therefore what happens when we start denying evidence for those priors if I'm in the planet which was just n of one and I guess are there differences there for if that was always the case versus I've just been plumped there and I've and now I have a memory in some sense of being a social creature which is now being denied to me I think yes there would be a huge difference um I would say if you just dropped there I mean it's a bit like being put into um solitary confinement I'm sure you will start thinking about the past or making up stories or about social interactions and so on and I was thinking also your nice point about being an MRI scanner all by yourself I suspect what most people do when they're an MRI scanner all by myself thinking about what on Earth is the experiment of doing this for sure I think that when I get x-rays actually I'm like why are they doing this with the blocks of lead and stuff yes if you'd never had social interactions I would I would think you would be well in the extreme I had a strange interaction long ago with Kristoff cooch talking about I was trying to talk about you know what's the function of Consciousness and I was putting the argument it's all social so my argument would be if you were never had interaction with other people you would not be conscious I think Cole has a similar yeah St yes and what was Kristoff's response to that know Consciousness is in these special neurons yeah yeah I don't think that one's got very far uh sorry Kristoff you're far more senior than me but I don't think that one's got very far and he lost that bet to David Chalmers I think he had to give him that case of wine yeah so so I think he lost um well that's a really interesting point let's let's go there then because as I said this is really where I want to go is this self-modeling yeah and I'm thinking again this is something that I've spoken to KL about and I've heard Carl mention which is that we need in some sense to be able to infer the agency of others or their self model to be able to apply to ourselves a self model um there's a wonderful paper I like shouting out papers here so that people can go read and also just to give credit where it's You by yakob Hoy um and I think his last name is Michael I'm not sure what his first name is and it's called why does anybody have a self it's anybody there two words and they talk about sort of cultural imitation as really being foundational to the development of a sense of agency in the sense I'm going to shape myself model on what I see as kind of optimal a IC behavior on the part of others so I guess the question here is how necessary is this capacity for inference of the agentic state of another or the mental states of another for our own formation of agency and selfhood I certainly think that selfhood depends on interacting with others though I think you

could get quite a long way without doing mentalizing it could be based on Behavior yes okay extent um but I'm one of the things that we quite talk about quite a lot in our book is reputation which is very important economically I mean The Economist I mean not um Adam Smith who invented econo economics talked about he didn't call it reputation he called I can't quite remember the esteem with which other people hold you in was was more important than wealth and what the main driving factor in human behavior is to get the esteem of others is to have a good reputation in other words though sometimes money can be a sort of n proxy for this sure and they're all sorts of interesting experiments on this because if you have a good reputation then people will trust you they will choose to work with you and they will inv invest money in you and all that sort of thing and then that you have a very interesting um arms race with the Free Riders because you want to have a good reputation but you want to cheat people at the same time and then you have to that's why being being anonymous is jly useful from this sort of competitive situation because no one you don't have a reputation that carries with you and um and it seems to me the reputation is an important aspect of the self and of course the reputation is in is not in you it's in the minds of others yes exactly but in some sense it then gets transferred onto you I mean I think it's quite classic literary Trope in some sense I've mentioned Luigi pandelo before I don't know why but he's come up again here he's got a well very peculiar book called unoo chento one no one 100,000 notion is is that these are different ways of modeling oneself self and what what he goes around doing in this book is protagonist is he he goes around trying to spot himself in mirrors so that he can perceive himself um not so he's trying to catch himself looking at himself not as others would perceive him but sort of in this raw moment of self perception and he eventually dissolves or goes a bit mental yeah it's that's one thing that came up into mind there's another far more folk or or popular culture thing there's a very interesting show on BB on the BBC right now um called the traitors uh which I highly recommend which is it's about sort of 20 people in a castle and three of them are traitors and 17 of them are so-called faithfuls and they have to kind of Guess Who's Who and if the traitors stay in they win all the money if the faithfuls stay in they win all there's some trash reality element to it but it's a very interesting psychological recursive thing where there's lots of triple quadruple bluffing going on so that's really that in in action and reputation yes plays a very big role in trustworthiness and you're watching that as the objective sort of Godlike figure and saying well how could you not see that this person is so malign um but you're not there in some sense okay that's that's interesting I guess a further question I had and this is actually something I would love to explore personally um because I

don't think anyone's done that much work on this um so I put this down as a sort of thing that I might want to explore in the PHD which is that we seemingly have homogeneity in self-modeling um so there are certain things that just come out of the literature who I'm thinking about people like Dan Dan Harvey and Shan Gallagher and um yob that it's in some way narrational in some ways it's uh epistemic it's somewhat coherent um again this goes all the way back to like people like Derek parfit um but it's and and that is homogeneous across us I was do you think that that is quite a thought I had is that that's rooted in the fact that we all engage in similar processes to arrive at that self-modeling to arrive at that self-conception namely inference about others does that seem reasonable yes I'm not sure I mean I would have thought that there's a I mean I'm not sure are you saying that because of our in inherent individual cognitive processes being similar to others we arrive at the same end result in some sense but it's a little bit more specific in the sense that uh let's say I'm a I'm a child and uh I watch my parents enact presumably agentic Behavior yeah my model of agentic behavior therefore starts to align with theirs yeah and what I do is I in according to act of inference I act in order to provide evidence for that prior and so you see this kind of um convergence over time on a self model of course there's going to be variety but these core features are present I would think that certainly has a role but I think there's also a role for sort of top down cultural effects so you're told this is what agents how agents behave right as I was saying you're told I mean even the very simple level you're told that people be you know when people are if people are eating a lot it means they're hungry I mean even at this simple level our behavior is determined by our desires is something that to some extent is top down imposed but I was thinking you were mentioning all these various people there's an interesting gayen straon yep objects to this narrative story yes he does in and I'm inclined to I I think there are different individual differences in this I have no idea I don't have a plan I never had a plan for my life as it were me neither yeah and I I might be a quite different person next week or and that was yeah so so yeah Gay's critique is very apt here um and here I guess he's he's kind of actually criticizing dennet because dennet is quite keen on talking about the linguistic form of this narrational sense of self yakob how's got a similar point there which is that maybe we can replace the word narrational with it's it's not as catchy but something like a predictive coherence or hierarchical constraints or something like that in the sense that the higher levels of your uh generative hierarchy are tracking lower levels and so your preferences right like I like ice cream is going to dictate my behavior in the moment that I'm more likely to go get ice cream and that feeds back and forward right I think that maybe we could just I don't know

what that term would be um but no I think it's a reasonable critique cool that that's interesting yeah let's um let's go to Consciousness then because it was something that you brought up um so there were two quotes that I'll read back to you if I may that you've said and I'm curious about what you think these are from papers I haven't just I was a completely different person then yeah yes yes yes that's very yeah very Derek parfit there are lots of you um in 1995 this is a while ago in a very I like the title of this paper Consciousness is for other people you wrote the primary function of Consciousness is to permit high level interactions with other conscious beings and then this is so that's about function yeah and then in another paper you wrote phenomenological Consciousness is necessary for taking an intentional stance towards other agents to do this we have to be able to treat ourselves as agents also I think we can just brack off that second sentence yeah yeah because we kind of touched upon that but there's there's one point where it's function is interactions and the second one is almost nature the necessity for uh intent the the necessity for Consciousness to take an intentional stance I don't know which order you want to take that in but it would be really interesting to unpick both of those well I think we've taken that a bit further I start with the first one that sure is so this is this study on metacognition with various people particularly Nick Shay and Cecilia Hayes MH and the idea is and it but it comes from originally from our the work with bador which is about people working together you know this study I'm sure so people are working together to detect a weak visual signal and we show that two people working together and coming up with a consensus answer are more accurate than the more accurate person working on their own in the right context at least and the reason that this happens is that they're able to tell each other how confident they are in their in their belief and on a tri B all basis you choose the answer of the more confident person so in other words it depends on them telling each other about their phenomenal state if you like why yeah I've come across as David charma's parrot on many times in this in this podcast and I'm going to do it again why why is that necessarily the case so why could it not ju why could it not just be an offline computation whereby we have two probability distributions that are just done completely like offline non-phenomenal they're just numbers um and you know you have a higher order agent or or one of them you know there's just a there's just a algorithm if choose the higher one and none of this is done online why do the lights need to be on um for that um for that successful Interac the information has to get from one mind to the other yes yes if if we presuppose that uh well what for adaptive behavior in this example yeah yes could that not be achieved also with offline Minds I guess is my point could that why do we need to invoke like intentionality qualia subjectivity all the fundamental

tenants of Consciousness yes I do I don't really quite understand so CH so Chalmers in 1995 speaks about how you can give me a function of Consciousness so whether it's uh like a global workspace or whether it's binding or whether it's this and but you can never tell me about how that function maps onto its phenomenal nature and why that function couldn't happen I think what's wrong with that is he's talking about the individual brain and I'm talking about interactions between two people and the key thing of the interaction is that they have to communicate with each other in this case confidence yes but could we not have something like an artificial agent which has Precision which could be a proxy for confidence and they're not conscious they just have done some mathematical computations the inverse entropy and this is my confidence right based on like you could have that with a large language model for example so you would have to have so your argument be if you had two artificial agents they could also get better than one working on their own precisely in this situation yeah there you go that that would be my contention not that I again that doesn't discount the that doesn't discount the putative role of Consciousness it just discounts or calls into question the necessity for Consciousness I guess right but I mean on the on the basis of that sort of experiment we went a bit further and started talking about do we actually learn from each other how to interpret the is in the states that we have um and again it becomes very in intentional and the in and I'm not sure if this is relevant or not but when we talk to each other about how confident we are we can actually then you get in competitive situations we can express more confidence than is actually justified in order to in this experiment was to become an advisor so people would want to take your advice because you seem to be more confident about it there that seems yeah yes my only concern I guess is that um there are these interesting Works coming out about sort of extended Trust In Mind Robot interactions or mind artificial agent interactions and if I have a belief that even a robot is extremely predictable and reliable and productive I can offload um a lot of my capacities to that robot and actually if we were in this kind of task I could very happily take its uh its interpretation of the events or its position as my own and I might receive great outcomes but I don't need to invoke in that any representational space within that robot does that in some ways not call into question even something like confidence metacognition whatever it is again I'm not I'm not calling into question the fact that it might be this it's more the necessity of it right um anyway it's an interesting it's an interesting little rabbit hole Consciousness does this we've g into you know everyone I've had on and we talk about Consciousness it it leads into these rabbit holes um we had an interesting discussion on Tuesday The Institute of philosophy because they're all obsessed with large

language models yes and they were talking about does it mean when you know does it understand what it's saying and it was interesting that they the presenter was finishing up by saying is there is meaning but it doesn't mean what it says so basically they decided that to go the the prop you had to have intentionality right which large language models don't currently have how much of this is sort of rooted in um John SS work Chinese room and sort of thing yeah excellent yeah yeah yeah it's good I like to see that 80s philosophy still has its role um is what I was sort of brought up on um excellent let's now it's not a jump because I I think the wonderful thing about your work is that although it seems diverse they all sort of trickle down in into these fundamental questions of agency and selfhood um I want to talk about schizophrenia yes um and maybe we can start with the a very interesting symptom which is delusions of control yes so you spoke uh you wrote very extensively on delusions of control so again people who are not schizophrenic or don't have a psychological psychiatric background might not know what I mean by delusional control so it might be worthwhile starting there just outlining what that phrase means so this is where the patient says it is not me doing causing these actions there's some external agent making me do these things and they could be quite trivial things like combing my hair or they could be more um dangerous things like attacking people or something but the idea is and this is interesting because there's a delusion typically is considered to be a belief so you can say that they believe that external agents are causing their actions but it seems more likely and early psychiatrists had more or less stated this is it's not so much that I believe that external agents are causing my actions it's I feel as if external agents are causing my actions it's not just a belief in the sort of philosophical sense is something stronger than that well that's interesting so it goes beyond the propositional in some sense and there is part of a whole flu of symptoms called passivity phenomena whereby you can say I'm it's not my emotion it's being implanted in me and the most perplexing one of all they're not my thoughts yes I wanted I wanted to get to that that that is which is De weird yeah very weird very it might be worth um distinguishing the uh delusions of control from something like an alien hand syndrome just so people don't that's right might be aware of this notion of a Wandering hand so the alien hand syndrome is typically caused by a brain lesion and that's where the hand starts moving by itself and the patient says I'm not controlling it or I don't have control over it it's doing these things that I don't want it to do but typically the patient doesn't really say there's some alien force right making it do these things and they try to stop it I mean the classic Strange Love in the film if you do I I actually haven't seen that I know I know I'm going to get shout to that in the comments um no pet say is playing amongst

many other characters as Doctor Strange Love who's an ex Scientific Advisor in a wheelchair and he keeps making these well it seems to have some strange overlap perhaps with Tourette's um I'm working on again I don't come at me I'm not saying they're the same but it seems like that um I've come up with this term called oppositional self-modeling which is a one level of some prior a one sort of lower level of generative hierarchy you kind of you have a habitual um action policy so you'll just do it right so there's got to be some part of your motor Loop where you will move your hand or you will do a tick but then you have a higher conceptual level where you really don't want to do it because you aren't the type of agent that does tick or does move your hand randomly so that that's what I mean by sort of the overlap there with Tourette's there's also utilization Behavior after frontal lesions where people make these habitual responses situations where they shouldn't they can no longer inhibit them okay interesting yes so in terms of delusions of control perhaps you could explain to people where the model that you proposed um for where this might come from at least some of the versions of it in terms of the inverse modeling and forward modeling yeah well I guess it goes back to helmoltz who noted that when we move our eyes obviously the image on the retina jumps about but we don't perceive it jumping about and the idea is that we use what So-Cal coroller discharge so that you can when you send the message to move your eye you also send a message to the visual system so saying there's going to be movement now but it's not caused by something happening in the outside world it's due to my own eye movement and that applies to all sorts of movements and more recently particular say with Daniel walit you have this idea of inverse and forward models so it says when I move my arm I actually have a forward model that tells me what what where my arm is going to be when the movement is finished and what sort of Sensations are going to be accompanying the movement and because I can predict them I can discount them yes so sort of sensory attenuation yeah and um this is the difference between a passive movement and an active movement so in a passive movement when somebody else moves my arm I feel all these Sensations because there's no discounting and that's why can't tickle yourself yes and the idea was that in schizophrenia perhaps something went wrong with this system of control so it literally was the case that when they moved their arm it felt as if somebody else was moving it yes they and we showed or Sarah Jane blore showed that they could tickle themselves oh really I didn't actually see I didn't see that I'm aware of her work I know you got you fre all did wonderful work in this in this domain maybe it's worth um unpicking exactly what that looks like technically it's a shame that I can't get up all of the nice diagrams because there are some really nice diagrams

but it might be worth um talking about so people might think um okay I want to move my hand to grip this mug yeah this I love Dante mug oddly enough um because I'm a bit of a Dante acolyte that's beside the point um and I can just I I just know what motor action to employ and I just do it yes I think you term that I think that might be the inverse problem what why why is that why is that why is that sort folk psychology notion not possible for you know cognitive creatures like us as far as I understand it the inverse problem is that whenever you you have a goal of Your Action yeah which is to reach that mug in the appropriate way but there are an infinite number of different ways in which the movement could proceed and the question is how do you decide which one to use yes and there are various arguments like do you minimize the energy requirement mhm or the distance or or I um Daniel walut and I don't know whether he still thinks this thinks what you minimizes the endpoint error there're all so that's the sort of argument about and also there always different muscles you can use and it's all very complicated and luckily we don't have to think about it we just do it you but underlying there is something underlying all that and certain kinds of lesions you can become extremely difficult to grasp things interesting so so so there are potentially infinite ways of uh choosing a suite of sensory uh sorry motor movements to achieve a goal so it's kind of computationally intractable to find a unique sequence or a singular sequence um however we can predict the outcome of Ro sequences now that's a different point yes that the critical idea with the inverse model is intractable to work out what you what's the best thing to do but with the forward model once you have given a set of instructions You can predict precisely where you're going to finish up okay excellent and so let's so that's so we've got the inverse problem the forward model now and if we then fold in what's going to be another kind of inverse type which is the efference copy yeah what's the efference what's the role of the efference copy here as far as I understand the efference copy is just a way of saying we can predict what Sensations are going to arise from this movement and that's the the thing that enables you to stop the image apparently moving about because you're you know what the movement is going to be and if the outcome is the real outcome matches the predicted outcome and everything is fine yes but if it in the case of schizophrenia the prediction is wrong so it doesn't match the outcome so we have to explain this um prediction error and one way of explaining the prediction error is saying somebody move my arm I see and this and does this require us in some ways to go to for insertion to say that yeah action some way many people believe that thought is thinking is an activity this goes right back to Lura I think in Russian psychology um but it's philosophically becomes very complicated because you can say um you have to say

I intended to have a thought yes and that's definitely the Buddhist might have a problem with that if if you're thinking about a problem you you would have some idea of what sort of outcome you want to have as you work your way through this problem and so they are there is room there for um you know metacognitive approach to thinking so you you could certainly say this I have not succeeded in my thinking task yes you say I think I have a good even when you begin you can say I have a good chance of achieving this thinking task and I guess might relate that to tip of the tongue phenomena where you say I can't remember at this moment but I'm pretty sure if I try a bit harder I should get there so these would be examples of putting thinking into a sort of active framework yes there are some very nice metacognitive models actually or or meta awareness models with an active inference so um there's one by L SED Smith 2021 which is very beautiful and these really rich Bay graphs with these hierarchical layers um how so I believe there are I believe there's good evidence that people with schizophrenia do not have major problems in motor control how is this accounted for oh yes well that's exactly right there's always a schizophrenic whenever you come up with a theory of schizophrenia they should be much worse than they actually are yeah yes yes but the um they do have they have eye movement abnormalities sort of smooth Pursuit and things like that which would be consistent with failing to predict properly the consequences of eye movements I there's a there some nice experiments from Germany you can do strain how do this work you can do experiments where you have to track a moving Target and this means that you your impression is that the target stays in the middle of your field of view and the background moves though it's the target that's moving and the background is stationary and you can so you obviously have to correct for the S of the moving background as it were and you can measure how much how successfully people make these Corrections and schizophrenic patients are less good at doing that so there are some subtle F sure that you can see yes interesting does this actually I I've got one more question before we go to delusions and hallucinations because I sense those are slightly different yeah um I mean this is in some sense a delusional belief um so there's some interesting stuff there's some very um there's some very adjacent or or actually overlapping work in active inference uh about agency and it's it's very much a offshoot of the uh comparator models and what we've been talking about so and and a lot of this is actually now been focused on depersonalization yes so and idea I really like is comes from George Dean he's got this idea of an alistic control model and what that means is that if I can predict the sensory outcomes of my actions very similar to what you were saying and I get a sort of non prediction error right I get the results that I expect I can

start to develop the inference that there was there is an agent in my generative model that does these things right and what happens is so let's say let's say now that that's an inference and I need to keep providing evidence for that inference to keep it high Precision let's say now let's say I'm in a torture chamber this is an example uh torture is a very reliable uh producer of symptoms of depersonalization right say I'm now getting tortured I have a prior that I am the type of agent that should be able to uh do things in to confirm evidence for my own uh model of the world but in the torture situation I might have a desired action but I'm not able to actually conf do it and therefore provide sensory evidence for it right and so what you start seeing here is the disentangling of that very inference that there is some some agent uh at the heart of my generative model does that how does that sound to you does it sound very much aligned with the way you would think about agency yes I no I think that's very interesting um I mean we there certainly I've been very interested in the feeling of being in control and there experiments on that um and I guess the torture situation is where you're totally not in control which we likely didn't normally experience yes um I'm just trying to think yes a lot this goe I mean it's a very basing story that the feeling in control of being in control depends not only on the intention to do something but also on the outcome yes exactly exactly yeah yeah I've been writing with car and others on so-called flow States yeah uh flow sta surfing rock climbing playing musical instrument and there although you don't have this Rich sense of self this epistemic agent model this propositional agent you do feel in control yes exactly and that is exactly as you say because you are you have this like alignment between your the predicted consequences of your action and the actual sensory outcomes in this very tight motor Loops um and I think it it's worth folding in this notion of inter receptive inference as well so this is people like annel Seth and Lisa Felman Barrett this notion that at the very core what um Thomas Metson go would call your minimal phenomenal self they've argued that this is rooted in inter successful interceptive inference yeah either through autonomic reflex or eating right some actual volitional action and I guess something else there is that torture would also uh deny the possibility for successful inter receptive inference if I'm not getting fed I am weakening I'm I I am unable that that prior that I need to have a certain level of glucose is not going anywhere um so I think what happens there is that the the the the agentic model itself dissolves rather than the homeostatic requirements yeah so there's there's there's some really beautiful work being done here um excellent before we go on to delusions and hallucinations uh um I wanted to ask I so my my S for is into Psychiatry are limited I I'm more Adept at cognitive science but I did read over the summer uh Rd Langs The Divided self I was curious

um he he he's famous for sort of instituting a more I would say empathetic uh and understanding approach to Psychiatry that these people are not just crazy right but they are what we would now say are actually Bas optimal just confirming priors that don't really correspond to what we would see as adaptive Behavior I was just curious about what your opinions are of whether you have any of R Lang and what he puts forward in the div divided self I I no I don't have a very high opinion of R Lang but then this is mostly due to interacting with other psychiatrists um and I I don't think I might have read The Divided s but I don't think I did the idea I mean putting it into that basian story is quite because that seems a very good idea that is to say they have the wrong prize but they're behaving I mean certainly all the stuff on delusions yes is the suggests that PE the original idea was that they're irrational yeah whatever that means whatever that means and that in other words they are not argue you know they're not appropriately arguing from the evidence and um all the experiments that ever been done sh they're extremely rational and they're probably more irrational than other people and as you say it's just that they start with the wrong priors yes there's something in basian modeling called the complete class theorem which is that any any action perception can be cast as Bas optimal just given a certain set of Prior they're slightly worrying yeah slightly worrying well but it accounts for things um it accounts for all I mean basically all phenomena because if you speak to co well actually you wouldn't go that far but you could cast it all as this kind of variational basian inference so it's all Bas optimal everything is BAS optimal um but I guess the the flip side of that the positive side of that is that maybe it does license a more humanistic approach to psychiatric patients in so far as there's this interesting argument I don't know if it's an argument there's an interesting point which is that maybe something like this going slightly away from schizophrenia but it's relevant something like PTSD or uh OCD or general anxiety disorder actually is could be quite adaptive in a certain ecological niche absolutely if I'm the war zone being you know having hyper salience um and being really you know having very high Precision over the likelihood of some sensory input and downweighting priors like in you know arguably in PTSD is actually really useful um but it's just not adaptive in modern you know calm London on leafy regions Park so how do do you think in your experience in Psychiatry this work has altered the way that we view psychiatric patients yes I think it has um I mean in psychiatry in reality of course it's very difficult because there aren't enough of them and there's no money and they can't find any drugs that work but I mean that's the how that really operates but going back to slightly earlier question people like gosh got his name there was the idea that the symptoms of schizophrenia not understandable

which is in another you know they have nothing to do with reality or our lives and I guess one of my AIMS in the sort of story story of delusionist control is to say they are understandable it literally does feel as if somebody was moving my arm and and how much would you say this has come from your it would you say that person yeah yeah do do you sense that this um empathy that you have in in the very literal sense of empathy that you can that it's understandable comes from direct work you know directly working with these people oh to some extent yes I mean in the when I was with the medical research Council my lab so all MH was in the middle of the acute psychiatric ward so I was interacting with them every day and it was quite good because they were actually very bored so they would come and knock on the door and say have you got another experiment for me to do yes I think I think uh that's where K started was in psychiatric rotations yeah yeah how how did you guys first meet if I may ask was it at UCL no we met at the um the psychotron unit at the hsih hospital oh W where we had a there was a project I think run by Peter little to basically start scanning patients with schizophrenia to test various theories and I if I remember rightly Carl was appointed because he had the psychiatric background to work on this project and that was in the very early days before we knew how to analyze brain Imaging data yes so he was given the task of analyzing the brain Imaging data and the rest is history as they say yeah quite quite what was it like uh what was your your most cited paper on ever yeah is the uh statistical parametric mapping yeah work that you did again we're going a little bit off piece but this is the beauty of this podcast like you know we can go wherever we want what was it like be again I don't know how SPM works I don't know how I'm not a brain scanner so maybe one day um or voxel base morphometry or any of these things but I'm just curious about what it was like no it was very exciting yeah exciting and I think what was wonderful about Carl is that I mean there lots of other people doing Imaging for the first time and they were tending to invent new ways of analyzing the data yes whereas Carl very sensibly said but there are all these statistical paradigms already there yeah these are what we should be using and that's yes I've always kind of wondered how Carl knew about them in the first first place um but I I sense that your work on uh forward modeling um should I I you know I was reading I had the joy of going not obviously through the entire catalog of your works because that would take me until next year but going through a couple of them again and some of them are sort of canonical I've read them before um and it's what struck me almost like reading Thomas metzinger I've said this before about metzinger how incredibly forward like in retrospect it just seems like you're speaking the same language as we're saying what we're saying now in active inference

Landscapes not quite but nearly not quite but well there's a lot of overlap and it's incredibly influential and I would say that for you and uh Sarah Jane and someone like Thomas metzinger it's it's really remarkable how um you know how much was how much how how in conversation were you with Carl when he was beginning to develop the free energy no all the time we shared an office oh nice so I have lots of passive smoking experience oh yeah he he he does like a cigarette fair enough I think he's I think he's warranted one and um have you given much thought I guess we are on the active inference podcast so I ought to ask something more specific I know we spoke before an email and you said you know this is not directly your line of work yeah but you were around for when the 2010 paper was released the 2009 paper was released uh I think a free you know 2005 might be the sort of Beginnings um what were your kind of experiences around then and and how did you contribute well I'm not quite sure yes I Carl has his habit of putting my name on papers from time to time and I guess I contribute to some extent by saying we need more independent references or something like that or or I sure sometimes change the wording a bit and know what I was particularly interested is I was very keen and it really goes right back to Richard Gregory and people like that the the approach to perception and the idea that we if our you know if if there's a prediction error we have to update our model world and I thought that was very exciting and wrote about that in my previous book and so I was particularly excited with active inference because that gave you the other side of the picture and when we get a prediction era we don't have to update our model of the world we can change the world instead isn't that amazing I thought that was wonderful I know it is wonderful what what are your thoughts so this is kind of what if anyone's familiar this is what we would call the low road to active inference yeah this is predictive coding uh Helm Holan perception and the like and action as action as inference I guess you could say yeah have you given any thought to the so-called high road and by that I mean dynamical stochastic systems and no that's too difficult for me I'm trying to learn the maths yeah you're not wrong you're not wrong it is rather tricky excellent um okay so I said we would talk about hallucinations and delusions I think delusions we've somewhat touched upon which is kind of this uh fixity of priors um that is not allowing for updating but I read you know I read kapur's work on um sort of the dope aberant salience model yeah I never thought that any of this stuff really accounted for hallucinations where where where do you the best of your knowledge do you think hallucinations might come from well I I with Paul fetch we read a paper which we're basically saying there's no fundamental difference between hallucinations and delusions okay if you take a basian approach yes yes because the

hallucinations are it's just about perception rather than beliefs but in both cases it's all about whether you update and where the evidence comes from but the thing you have to remember about schizophrenia is these hallucinations are predominantly auditory and they're predominantly hearing voices and One symptom is so all thought be podcast or no yes you can you I hear my thoughts as if they're being spoken aloud um and you can even have a sort of very similar story with the forward model and there's work on this so that when you speak you hear your own voice but you suppress it and there's some evidence that when patients speak that again the suppression system seems to go wrong in relation to speaking as well and you have inner speech and all these sorts of things so the one idea is that they're perceiving their own inner speech as if it was from an external source which is not that far away from a delusion of control yes I sometimes how true is that if when one meditates for example if one is doing vasna um it does seem like those thoughts are coming from not from you so this might be looping back into our sort of humanistic approach to psychiatry all so don't tell that to your psychiatrist yeah yes yeah yeah yeah I won't I it's true at least and and the B you know there's this these very odd overlaps between Buddhism and Psychiatry and ego death and all of these kind of things do you think that this this notion that self talk let's say let's stick with that because it's very interesting is self talk from a Buddhist from a very raw Buddhist perspective that doesn't get off the ground because there is no self these thoughts are just popping into awareness yeah and it's it's all process it's process ontology it's not there's no substance there it's it's coming and going it's just the field of awareness so is it just an Adaptive tool for us to presume or or have this illusion that it's self-generated I guess that depends what you mean by self-generated I mean well not self-generated but it's what does self talk mean then that maybe that's because it's yes I mean clearly it's not I'm not intentionally saying these things per say but I'm perceiving them as belonging to me yes they're not coming from outside is yeah sure but what's in this in this way we have to kind of ask well what is inside and outside yeah so so well that's I mean that's a common one I mean one quite dominant theory of schizophrenia they've lost the boundary between inside and outside that's really interesting they've lost their Markov blanket yes if you yeah yes but I guess I guess the reality is you never lose a Markov blanket I mean this to be honest it it points to quite it points to theoretical difficulties in Act of inference which is something that's been brought up in this podcast before and in in other works which is where do you even cast the Markov blanket and what is it to say that you have internal States and external States really what those actually mean um and I give credit to Maxwell RAM for pointing this out to me is

that internal you can just say are tracking and external are tracked but that said you have someone like Andy Clark who might say well in my tracking mechanism I might have my telephone or my notepad and then you have job how and says no no no the evidential boundary is the mind and even the body is being predicted yeah does any of those uh dichotomies or debates inform something like um what we've been talking about self talk and uh delusions of control well self talk inner speech people used to call it yes and that is and again the Russian psychologists I mean vyty I think but and there's some evidence for this that you know young children particularly doing difficult tasks what we might call frontal tasks you actually talk aloud I must now do this and I must now do this and this becomes internalized so you're do that internally and I think for some difficult tasks and things like mental arithmetic there is indeed in the speech which is being used to control what you're doing interesting and um some schizophrenic hallucinations where the where they hear a voice commenting on their action so now he's going to do this and then he why is he doing that and it's all very weird what they but you can see a sort of thread coming through this I mean that's what I would call self talk and self is not important well I wonder whether there is some uh some space for a predictive model here yeah in so far as if we're going to talk about what we meant by self self-generated an agency we were talking about uh deploying some prediction and having in association with that prediction uh not just I don't just deploy an action policy I also uh precise predictions about the sensory outcomes I'm expecting to receive and perhaps we can talk about self talk in a similar way which is that at the Inception of my self talk let's say I'm predicting uh the Sens or sensory in uh Speech marks consequences of an action policy of saying this at $t+1$ one yes and then I receive that back and that seems self-generated yeah yeah and I guess if that mechanism itself is disrupted through synaptic problems Precision waiting problems then that is not only just going to be motor action but also mental action yeah is that appropriate absolutely yeah maybe I should be a psychiatrist mental action is a good term yeah yeah mental action is a very good term well I mean all of this um it's it's kind of become very clear to me that all such a plethora of behavior in in active inference is mental action I mean the very deployment of precision uh um over what now you know we're using these basian graphs these So-Cal pom DP schemes so you have your sensory expectations your C Matrix your uh State transitions your B Matrix these are deploying precisions over those are all mental actions yeah um so okay excellent okay that's that's okay interesting okay that's very that's very cool um I was wondering so there's a uh I believe he was a philosopher called Lou sass or maybe he was a psychologist and he wrote a book called

madness and modernity he yes was he more sort of modernism it might be it might be it's been a while since I read it yeah um well please please educate me because I'm yeah I mean I read it too but it was a very long time ago um so I mean it's a removal of sort of structures and Bas um um standard procedures and yeah stream stuff um and how does that feed into schizophrenia schizoph is sort of uh the iology of schizophrenia somewhat linked to having an unstable lifestyle um yes I mean there's an interest this is very um unlikely to be true but there's interesting stuff on volatility so with when the when the environment is volatile you should have a faster learning rate yes and I think that somehow interacts with the dopamine system okay so you could and there is some evidence that children brought up in very volatile environments are more at risk and one wonders whether and then you've got this doine aspect coming in as well if we've got too much damine so you get the there's that sort of story and modernism in the sense is increasing the volatility because you no longer rely on um I see cultural Givens you know currently an interesting idea in relation schizophrenia which is bringing the cultural thing is that most of our beliefs at this high level are constrained by interactions with other people and culture more generally and you could argue that what's odd about what the reason that schizophrenic beliefs don't update is because they're no longer constrained by this high level cultural if what other people say ah so these are yes your work is all coalescing exactly that's cool that's nice yeah that's very yes it's interesting isn't it because we think of ourselves as these we live in a rather individualistic Society where we think of ourselves as these atomist IC beings and we're not and we're not that's terrifying um good excellent so so the book came out last year yes what's on the what's on the horizon for Chris FFF boa FFF the fffs well um we're in our 80s now um you wouldn't be able to tell um we haven't got a major project in line um we have a quite exciting interaction with our friends in orus in Denmark yes where they have uploaded all our papers and are trying to construct a large language model interesting who we really are interesting uh is this Andreas well no it's in his his yeah yeah yeah yeah because you have a you have a partnership with him or is that still Ono we visit every year yes yes excellent he's great I think he might be coming on um yes you should certainly talk to him yeah no I'm extremely interested in this work Chris This was um this was a delight um it's been a delight on my part to go back and read uh your texts I mean I did finished my Master's just background about myself in case you didn't I finished my masters at UCL in experimental psychology last year and I would say sort of your presence and K's presence sort of lingers in a positive way not in a sort of ghoulish way through the corridors um so thank you y my pleasure it's been an absolute pleasure all right excellent